1. Project Title

Banking Risk & Fraud Analytics Dashboard: Customer Risk Profiling & Fraud Intelligence System

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2. Abstract

Financial fraud has become a major challenge for banking institutions due to the rapid growth of digital transactions. With millions of transactions processed every day, identifying fraudulent activities manually or through traditional rule-based systems has become increasingly difficult.

The **Banking Risk & Fraud Analytics Dashboard** is a data analytics project developed to analyze large-scale financial transactions, identify fraud patterns, evaluate customer risk, and provide actionable business recommendations.

The project uses the **PaySim Synthetic Financial Dataset containing more than 6 million transactions** and applies data analytics techniques using Python and Power BI. The analysis focuses on transaction behaviour, fraud concentration, customer risk profiling, balance anomaly detection, and fraud impact measurement.

Unlike traditional fraud dashboards that only display fraud counts and losses, this project works as a **fraud intelligence system** by analyzing transactions at multiple levels:

- Transaction-level risk analysis
- Customer-level risk profiling
- Account behaviour monitoring
- Business impact analysis

The final interactive Power BI dashboard enables financial institutions to identify suspicious activities, investigate high-risk transactions, and implement proactive fraud prevention strategies.

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3. Introduction

The banking industry generates massive amounts of transaction data every day. While digital payments provide convenience, they also increase the risk of financial fraud.

Fraudsters use different techniques such as unauthorized transfers, account manipulation, and suspicious cash withdrawals to exploit financial systems. Traditional fraud detection approaches often depend on fixed rules, which may fail to identify complex patterns hidden within large datasets.

Data analytics provides an effective solution by analyzing historical transaction data, identifying unusual behaviour, and generating risk indicators.

This project focuses on transforming raw banking transaction data into meaningful insights using Python-based analysis and Power BI visualization.

The objective is not only to identify fraudulent transactions but also to understand:

- Why fraud occurs
- Which transaction types are risky
- Which customers require monitoring
- How banks can prevent future fraud

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4. Business Problem

Banks face several challenges while detecting and preventing financial fraud:

- Millions of transactions make manual investigation difficult.
- Fraud transactions represent a small percentage of total transactions, making detection challenging.
- High-value fraudulent transactions cause significant financial losses.
- Repeated suspicious behaviour from certain accounts may go unnoticed.
- Traditional systems may identify fraud after the damage has occurred.

The project aims to develop an analytics solution that helps banks:

- Detect suspicious transaction patterns.
- Identify high-risk customers.
- Analyze fraud impact.
- Support proactive fraud prevention decisions.

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5. Project Objectives

Fraud Detection Analysis

- Identify fraudulent transactions from historical banking data.
- Analyze fraud patterns across transaction categories.
- Understand characteristics of fraudulent activities.

Customer Risk Assessment

- Identify customers involved in suspicious activities.
- Analyze customer transaction behaviour.
- Categorize customers based on risk level.

Transaction Risk Monitoring

- Identify high-value risky transactions.
- Detect unusual transaction behaviour.
- Analyze balance inconsistencies.

Business Intelligence

- Build interactive Power BI dashboards.
- Generate meaningful fraud insights.
- Provide recommendations for improving fraud prevention.

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6. Dataset Information

Dataset Name

PaySim Synthetic Financial Transactions Dataset

Dataset Size

- 6.3 Million+ banking transactions
- Multiple transaction and customer attributes

Dataset Description

The dataset simulates real-world financial transactions and contains information related to:

- Transaction details
- Sender accounts

- Receiver accounts
- Account balances
- Fraud indicators

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7. Dataset Features

Feature	Description
step	Time period of transaction
type	Transaction category
amount	Transaction value
nameOrig	Sender account ID
oldbalanceOrg	Sender balance before transaction
newbalanceOrig	Sender balance after transaction
nameDest	Receiver account ID
oldbalanceDest	Receiver balance before transaction
newbalanceDest	Receiver balance after transaction
isFraud	Fraud indicator
isFlaggedFraud	Rule-based fraud flag

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8. Tools & Technologies Used

Programming & Data Processing

- Python
- Pandas
- NumPy

Data Visualization

- Power BI
- Matplotlib

- Seaborn

Analytical Techniques

- Exploratory Data Analysis
- Fraud Pattern Analysis
- Customer Risk Segmentation
- Anomaly Detection
- Business Intelligence Reporting

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9. Data Cleaning & Preparation

Before performing analysis, the dataset was cleaned and transformed.

Data Cleaning Steps

Duplicate Removal

Duplicate records were identified and removed to improve data quality.

Missing Value Handling

Missing values were analyzed and handled to maintain dataset consistency.

Data Type Optimization

Data types were converted for efficient analysis.

Examples:

- Amount → Numeric format
- Fraud indicators → Boolean format
- Transaction types → Category format

Dataset Optimization

Since the dataset contained more than 6 million records, optimization techniques were applied to improve Power BI performance.

Steps included:

- Removing unnecessary columns
- Creating analytical datasets
- Optimizing data model relationships

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10. Feature Engineering

Additional metrics were created to improve fraud analysis.

Fraud Rate %

Measures the percentage of fraudulent transactions.

Formula:

Fraud Rate = (Fraud Transactions / Total Transactions) × 100

Fraud Loss

Calculates the total financial impact caused by fraudulent transactions.

Formula:

Fraud Loss = Sum(Transaction Amount where isFraud = 1)

Customer Risk Classification

Customers were categorized based on:

- Fraud involvement
- Transaction frequency
- Transaction value
- Suspicious behaviour patterns

Risk Categories:

Low Risk

Customers showing normal transaction behaviour.

Medium Risk

Customers showing occasional suspicious activity.

High Risk

Customers involved in multiple suspicious transactions.

Balance Anomaly Detection

Balance behaviour was analyzed by comparing:

- Previous account balance
- Transaction amount
- Updated account balance

This helps identify:

- Unexpected balance changes
- Suspicious money movement
- Possible account manipulation

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11. What Makes This Project Different?

The major difference between this project and traditional fraud dashboards is that it focuses on **fraud intelligence rather than only fraud reporting**.

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1. Goes Beyond Basic Fraud Detection

Most fraud dashboards only display:

- Total fraud cases
- Fraud percentage
- Fraud amount

This project provides deeper analysis by identifying:

- Why fraud occurs
- Where fraud occurs
- Which accounts are risky
- What actions banks should take

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2. Customer-Level Risk Profiling

Unlike traditional transaction-only analysis, this project evaluates customers and accounts.

It identifies high-risk customers based on:

- Transaction frequency
- Transaction value
- Fraud history
- Suspicious activity patterns

This helps banks move from:

"Which transaction is fraudulent?"

to:

"Which customers/accounts are creating future risk?"

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3. Balance Anomaly Detection

The project uses account balance behaviour as an additional fraud indicator.

By analyzing balance changes before and after transactions, suspicious activities can be identified.

Example:

Unexpected balance reduction after a transaction may indicate abnormal financial behaviour.

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4. Investigation-Oriented Dashboard

The dashboard is designed as a fraud investigation tool rather than only a visualization report.

It helps analysts answer:

- Which accounts are suspicious?
- Which transactions need investigation?
- Which transaction types have higher fraud risk?
- What is the financial impact?

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5. Business-Focused Approach

The project converts analytical findings into practical business actions.

Recommendations include:

- Real-time fraud monitoring
- Automated risk alerts
- Multi-factor authentication
- Suspicious account tracking

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6. Large-Scale Data Handling

The project works with more than 6 million transactions, representing a real-world banking analytics challenge.

The project demonstrates:

- Large dataset processing
- Data optimization
- Business metric creation
- Interactive reporting

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7. Multi-Level Fraud Analysis Framework

The project analyzes fraud at three different levels:

Transaction Level

Analyzes:

- Fraudulent transactions
- Transaction amount
- Transaction type

Customer Level

Analyzes:

- Customer risk
- Account behaviour

- Suspicious patterns

Business Level

Analyzes:

- Fraud losses
- Risk trends
- Prevention strategies

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12. Dashboard Architecture

The Power BI dashboard consists of seven analytical pages.

Page 1: Executive Overview

Objective

Provide an overall view of banking fraud performance.

KPI Cards

Total Transactions

Total number of transactions analyzed.

Total Customers

Number of unique customers involved.

Fraud Cases

Total fraudulent transactions detected.

Fraud Rate %

Percentage of fraudulent transactions.

Total Transaction Amount

Total monetary value processed.

Fraud Loss

Financial impact caused by fraud.

Business Questions Answered

- How much fraud occurred?
- What is the financial impact?
- What percentage of transactions are risky?

Page 2: Customer Risk Analysis

Objective

Identify customers requiring additional monitoring.

Visualizations

Fraud Distribution Donut Chart

Displays:

- Genuine transactions
- Fraud transactions

High-Risk Customer Analysis

Identifies accounts with:

- Multiple fraud transactions
- High transaction values
- Suspicious patterns

Insights

- Some accounts contribute significantly to fraud activities.
- High-risk customers require additional verification.

Page 3: Fraud Investigation

Objective

Perform detailed analysis of suspicious transactions.

Visuals

Top Fraudulent Senders

Shows accounts involved in maximum fraudulent activity.

Transaction Amount Analysis

Identifies high-value suspicious transactions.

Balance Anomaly Scatter Plot

Analyzes relationship between:

- Old balance
- New balance

Suspicious Transaction Table

Contains:

- Transaction type
- Sender ID
- Receiver ID
- Amount
- Fraud status

Page 4: Customer Behaviour & Risk Profiling

Objective

Understand customer transaction patterns.

Analysis Areas

- Transaction frequency
- Average transaction amount

- Account activity
- Risk behaviour

Insights

Customers showing unusual transaction behaviour can be flagged for investigation.

Page 5: Fraud Trends & Pattern Detection

Objective

Identify fraud trends and common fraud patterns.

Visualizations

Fraud Trend Analysis

Shows fraud activity over time.

Fraud by Transaction Type

Analyzes fraud concentration in:

- TRANSFER
- CASH_OUT
- PAYMENT
- CASH_IN

Findings

- Fraud is mainly concentrated in TRANSFER and CASH_OUT transactions.
- High-value transactions contribute significantly to fraud losses.

Page 6: Advanced Risk Metrics

Objective

Create additional fraud intelligence indicators.

Metrics

Transaction Risk Score

Based on:

- Amount
- Transaction type
- Customer behaviour

Fraud Severity Score

Measures impact of fraud cases.

Factors:

- Transaction value
- Financial loss
- Account behaviour

Account Risk Indicator

Identifies suspicious accounts.

Page 7: Business Insights & Recommendations

Key Findings

- Fraud is concentrated in specific transaction types.
- TRANSFER and CASH_OUT transactions represent higher risk.
- High-value fraudulent transactions contribute major financial losses.
- Some accounts repeatedly appear in suspicious activities.
- Balance anomalies can indicate fraudulent behaviour.

Business Recommendations

1. Real-Time Fraud Monitoring

Implement real-time monitoring for:

- High-value transactions
- Risky transaction types
- Suspicious accounts

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2. Automated Risk Alerts

Generate alerts for:

- Multiple suspicious transactions
- Unusual transaction behaviour
- Abnormal balance changes

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3. Multi-Factor Authentication

Strengthen authentication for:

- High-risk customers
- Large-value transactions

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4. Balance Anomaly Detection

Implement automated rules to identify suspicious balance movements.

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5. Continuous Fraud Model Improvement

Regularly update fraud detection strategies based on changing fraud patterns.

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13. Challenges Faced

Handling Large Dataset

The dataset contained more than 6 million rows, creating performance issues.

Solution:

- Optimized data structure.
- Removed unnecessary fields.

- Created efficient data models.

Fraud Data Imbalance

Fraud cases were significantly smaller compared to genuine transactions.

Solution:

Focused on:

- Risk analysis
- Pattern detection
- Customer profiling

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14. Skills Demonstrated

Technical Skills

- Python
- Pandas
- NumPy
- SQL concepts
- Power BI
- DAX
- Data Visualization

Analytical Skills

- Fraud Investigation
- Risk Assessment
- Pattern Recognition
- Business Analysis
- Decision Making

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15. Future Enhancements

Future improvements include:

- Machine Learning fraud prediction models
- Real-time fraud detection system
- AI-based anomaly detection
- Automated investigation workflows
- Customer behaviour prediction

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16. Conclusion

The Banking Risk & Fraud Analytics Dashboard demonstrates how data analytics can transform millions of financial transactions into meaningful business intelligence.

The project goes beyond traditional fraud reporting by combining transaction analysis, customer risk profiling, anomaly detection, and business recommendations.

By providing a complete fraud intelligence framework, the dashboard helps financial institutions identify risks earlier, reduce financial losses, and make data-driven fraud prevention decisions.